STAT/MA 41600
In-Class Problem Set #36: November 11, 2026
Solutions by Mark Daniel Ward

Problem Set 36 Answers

In this problem set, Z always represents a standard normal random variable.

1a. Let X denote the amount of sugar in 5 cans of soda. Then X has a Normal distribution with mean $(5)(39) = 195$ grams and standard deviation $\sqrt{(5)(2^2)} = \sqrt{20}$ grams. We compute $P(X > 197) = P((X - 195)/\sqrt{20} > (197 - 195)/\sqrt{20}) = P(Z > 0.45) = 1 - P(Z \leq 0.45) = 1 - 0.6736 = 0.3264$.

1b. Let X denote the amount of sugar in 12 cans of soda. Then X has a Normal distribution with mean $(12)(39) = 468$ grams and standard deviation $\sqrt{(12)(2^2)} = \sqrt{48}$ grams. We compute $P(460 < X < 470) = P((460 - 468)/\sqrt{48} < (X - 468)/\sqrt{48} < (470 - 468)/\sqrt{48}) = P(-1.15 < Z < 0.29) = P(Z < 0.29) - P(Z \leq -1.15) = P(Z < 0.29) - P(Z \geq 1.15) = P(Z < 0.29) - (1 - P(Z < 1.15)) = 0.6141 - (1 - 0.8749) = 0.4890$.

2. Let X denote the weight of 10 cats. Then X has a Normal distribution with mean $(10)(12) = 120$ pounds and standard deviation $\sqrt{(10)(5^2)} = \sqrt{250}$ pounds.

2a. We compute $P(X > 125) = P((X - 120)/\sqrt{250} > (125 - 120)/\sqrt{250}) = P(Z > 0.32) = 1 - P(Z \leq 0.32) = 1 - 0.6255 = 0.3745$.

2b. We compute $P(112 < X < 128) = P((112 - 120)/\sqrt{250} < (X - 120)/\sqrt{250} < (128 - 120)/\sqrt{250}) = P(-0.51 < Z < 0.51) = P(Z < 0.51) - P(Z \leq -0.51) = P(Z < 0.51) - P(Z \geq 0.51) = P(Z < 0.51) - (1 - P(Z < 0.51)) = 2P(Z < 0.51) - 1 = (2)(0.6950) - 1 = 0.3900$.

2c. There may not be any fat cats in the group. For instance, if each cat weighs 14.5 pounds, then their total weight is 145 pounds, but none of them are fat cats. Other small variations of their weights are possible, with weight 145 pounds altogether, but no fat cats.

3. The amount of sugar in 150 cans of soda is a Normal random variable with mean $(150)(39)$ grams and standard deviation $(2)(\sqrt{150})$ grams. Converting grams of sugar to pounds, we see that the sugar in 150 cans of soda is a Normal random variable with mean $(150)(39)(2.2)/(1000)$ pounds and standard deviation $(2)(2.2)/(1000)(\sqrt{150})$ pounds.

Let X denote the weight of the sugar (in pounds) minus the weight of a cat (also in pounds). Then X is a Normal random variable with mean $(150)(39)(2.2)/(1000) - 12 = 0.87$ pounds and standard deviation $\sqrt{((2)(2.2)/(1000))^2(150) + 5^2} = 5.000290 \approx 5$ pounds. We have $P(X > 0) = P((X - 0.87)/5 > (0 - 0.87)/5) = P(Z > -0.17) = P(Z < 0.17) = 0.5675$.

4. The total length of the 1075 blades of grass (in inches) is a Normal random variable X with mean $(1075)(4) = 4300$ inches and standard deviation $(0.75)(\sqrt{1075}) = 24.59$ inches.

The probability that the blades of grass are (altogether) longer than a football field is $P(X > 4320) = P((X - 4300)/(24.59) > (4320 - 4300)/(24.59)) = P(Z > 0.81) = 1 - P(Z \leq 0.81) = 1 - 0.7910 = 0.2090$.

5. The total height of 15 blades of grass is a Normal random variable with mean $(15)(4)$ inches and standard deviation $(0.75)(\sqrt{15})$ inches.

Let X denote the height of the student (in inches) minus the total height of the blades of grass (in inches). Then X is a Normal random variable with mean $64 - (15)(4) = 4$ inches and standard deviation $\sqrt{(4.8)^2 + (0.75)^2(15)} = 5.61$ inches. We have $P(X > 0) = P((X - 4)/5.61 > (0 - 4)/5.61) = P(Z > -0.71) = P(Z < 0.71) = 0.7611$.